

Executive Summary

Final Model

XGBoost - weighted selected by predefined F1 rule after leakage-safe temporal validation.
Threshold 0.500; ROC-AUC 0.594; PR-AUC 0.082; F1 0.142; recall 0.361; precision 0.088.

Ranking Value

Top 10% ranked impressions achieved 9.25% CTR, 1.50x average lift, and captured 15.0% of holdout clicks.

SMOTENC Evidence

SMOTENC reduced false negatives by 21.7% versus class weighting in the controlled sample, but increased training rows by 14.1% and lowered PR-AUC.

Interpretation

Performance is moderate and calibration is imperfect, so scores are best used as click-propensity rankings rather than literal click probabilities.

Data Quality Summary

Data check	Result
Training observations	463291
Test observations	128858
Clicks	31331
Non-clicks	431960
Click rate	6.76%
Duplicate rows	0
Total missing cells	563955
Unique products	10
Unique campaigns	10
Unique webpages	9
Date range	2017-07-02 00:00:00 to 2017-07-07 23:59:00

Missing Value Summary

field	missing_pct
product_category_2	78.97
city_development_index	27.01
gender	3.94
user_group_id	3.94
age_level	3.94
user_depth	3.94

Only fields with non-zero missingness are shown.

Model Comparison ? Untouched Final Holdout

	model	threshold	roc_auc	pr_auc	f1	precision	recall	tn	fp	fn	tp
	XGBoost - weighted	0.5000	0.5935	0.0819	0.1417	0.0881	0.3611	50626	16439	2812	1589
	XGBoost - weighted - tune	0.5000	0.5935	0.0819	0.1417	0.0881	0.3611	50626	16439	2812	1589
Logistic Regression	- balanced - tune	0.4650	0.5955	0.0817	0.1395	0.0831	0.4329	46051	21014	2496	1905
	Logistic Regression - balanced	0.5000	0.5955	0.0817	0.1359	0.0904	0.2740	54928	12137	3195	1206
Random Forest	- balanced - tune	0.4650	0.5901	0.0810	0.1344	0.0874	0.2904	53720	13345	3123	1278
	Global CTR probability	0.0704	0.5000	0.0616	0.1160	0.0616	1.00	0	67065	0	4401
	Random Forest - balanced	0.5000	0.5901	0.0810	0.1049	0.0990	0.1116	62594	4471	3910	491
	Always non-click baseline	0.5000	0.5000	0.0616	0.0000	0.0000	0.0000	67065	0	4401	0

Thresholds are selected on the middle temporal split, not on this final holdout.

SMOTENC vs Class Weighting Experiment

strategy	training_rows_after_sampling	precision	recall	f1	pr_auc	false_negatives	training_time_sec	threshold
Class weighting	50000	0.0713	0.5183	0.1253	0.0762	2120	9.55	0.4700
SMOTENC	57038	0.0697	0.6230	0.1254	0.0725	1659	14.92	0.1540

SMOTENC is applied before one-hot encoding. It reduces false negatives but increases training size/time and does not improve PR-AUC.

Bootstrap Confidence Intervals

	model	f1	f1_ci_low	f1_ci_high	pr_auc	pr_auc_ci_low	pr_auc_ci_high
Leading model	Logistic Regression - balanced	0.1395	0.1345	0.1439	0.0817	0.0785	0.0853
	Random Forest - balanced	0.1344	0.1274	0.1402	0.0810	0.0773	0.0847
	XGBoost - weighted	0.1417	0.1352	0.1479	0.0819	0.0781	0.0857

Leading model intervals overlap, so XGBoost should not be treated as decisively superior from F1 alone.

Feature Ablation ? Incremental Value

	model	threshold	roc_auc	pr_auc	f1	precision	recall
Baseline	raw + temporal - tuned	0.4800	0.5301	0.0676	0.1193	0.0660	0.6199
+ Interaction	on history counts - tuned	0.5150	0.5759	0.0762	0.1334	0.0782	0.4549
+ Leakage-safe	safe historical CTR - tuned	0.4650	0.5955	0.0817	0.1395	0.0831	0.4329

Interaction history and leakage-safe CTR features add predictive value over raw + temporal features.

Lift / Gains by Prediction Decile

decile	impressions	clicks	actual_ctr	avg_score	lift_vs_baseline	clicks_captured	relative_clicks_captured
Top 0-10%	7147	661	0.0925	0.5718	1.50	0.1502	0.1502
Top 10-20%	7147	616	0.0862	0.5291	1.40	0.1400	0.2902
Top 20-30%	7146	568	0.0795	0.5009	1.29	0.1291	0.4192
Top 30-40%	7147	479	0.0670	0.4781	1.09	0.1088	0.5281
Top 40-50%	7146	443	0.0620	0.4547	1.01	0.1007	0.6287
Top 50-60%	7147	399	0.0558	0.4309	0.9066	0.0907	0.7194
Top 60-70%	7146	379	0.0530	0.4051	0.8612	0.0861	0.8055
Top 70-80%	7147	351	0.0491	0.3760	0.7975	0.0798	0.8853
Top 80-90%	7146	278	0.0389	0.3338	0.6317	0.0632	0.9484
Top 90-100%	7147	227	0.0318	0.2352	0.5158	0.0516	1.00

Top-ranked impressions concentrate clicks, which is central for ad-serving ranking decisions.

Precision@K and Recall@K

targeted_impressions	impressions_served	precision_at_k	recall_at_k	lift
Top 1%	714	0.1036	0.0168	1.68
Top 5%	3573	0.0929	0.0754	1.51
Top 10%	7146	0.0925	0.1502	1.50
Top 20%	14293	0.0893	0.2902	1.45

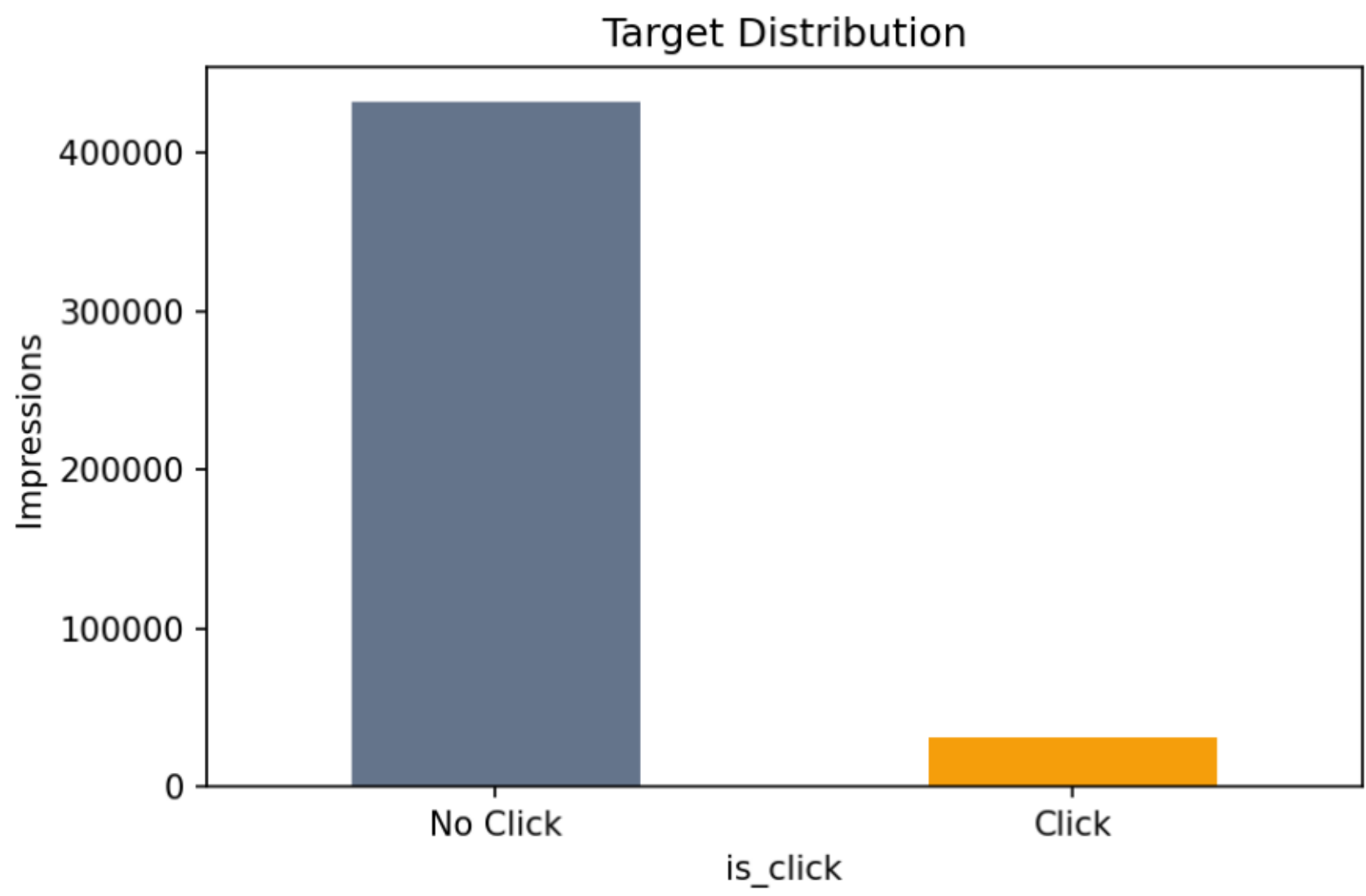
Budget-constrained serving view: how many clicks are captured when targeting only top-ranked impressions.

Business Value Threshold Scenarios

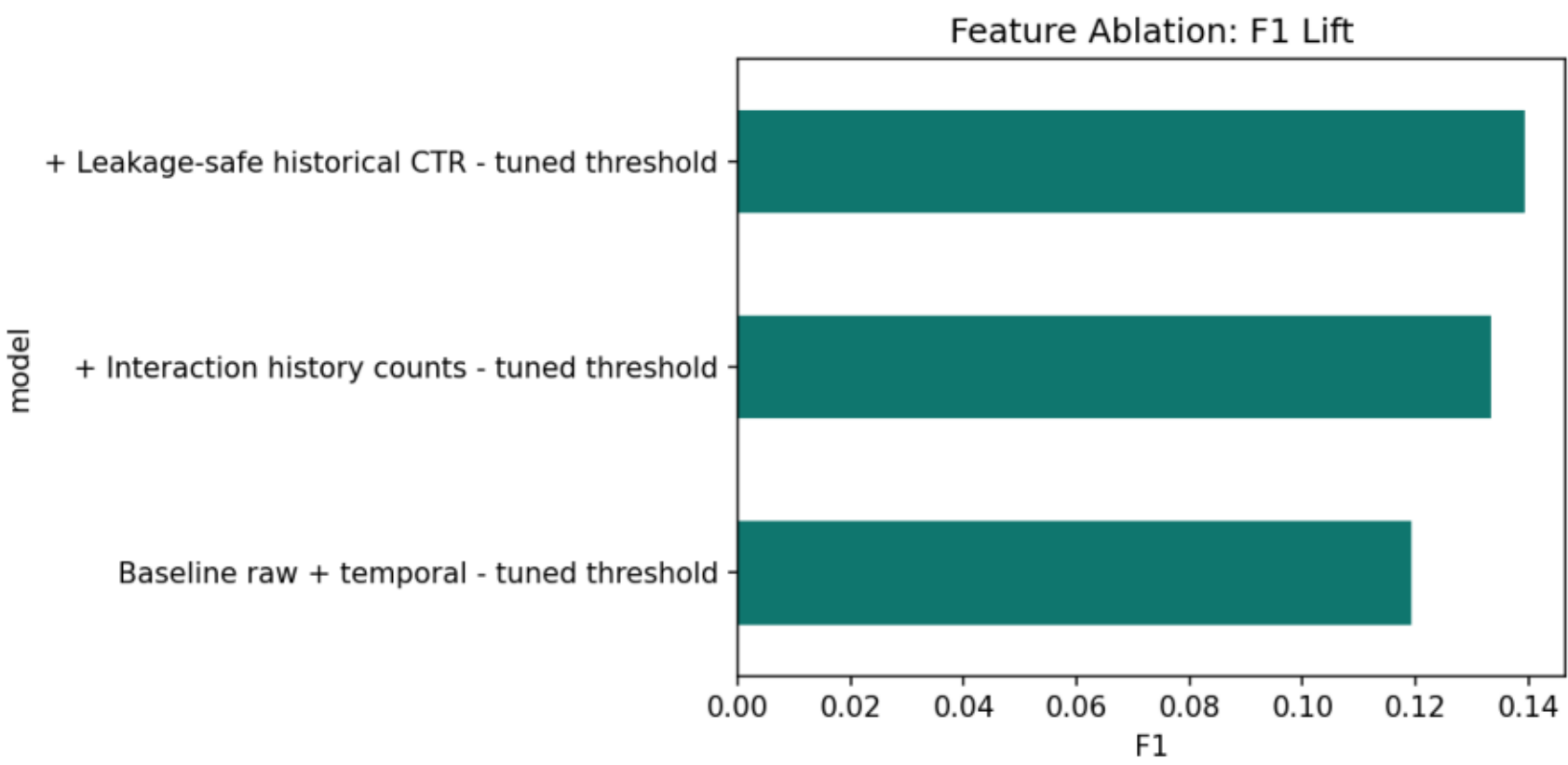
		scenario	value_per_click	cost_per_impression	optimal_threshold	served_impressions	clicks_captured	clicks_missed	precision	recall	expected_utility	utility_per_1k_impressions
Low click value / cheap	Base ad economy		1.00	0.0500	0.3910	50234	3557	844	0.0708	0.8082	1,045.30	14.63
			2.00	0.0500	0.1830	70241	4381	20	0.0624	0.9955	5,249.95	73.46
		High click value	5.00	0.0500	0.1340	70872	4396	5	0.0620	0.9989	18,436.40	257.97
		Expensive media	2.00	0.1500	0.4900	20842	1804	2597	0.0866	0.4099	481.70	6.74

Scenario assumptions only: utility = clicks captured x value per click - served impressions x cost per impression.

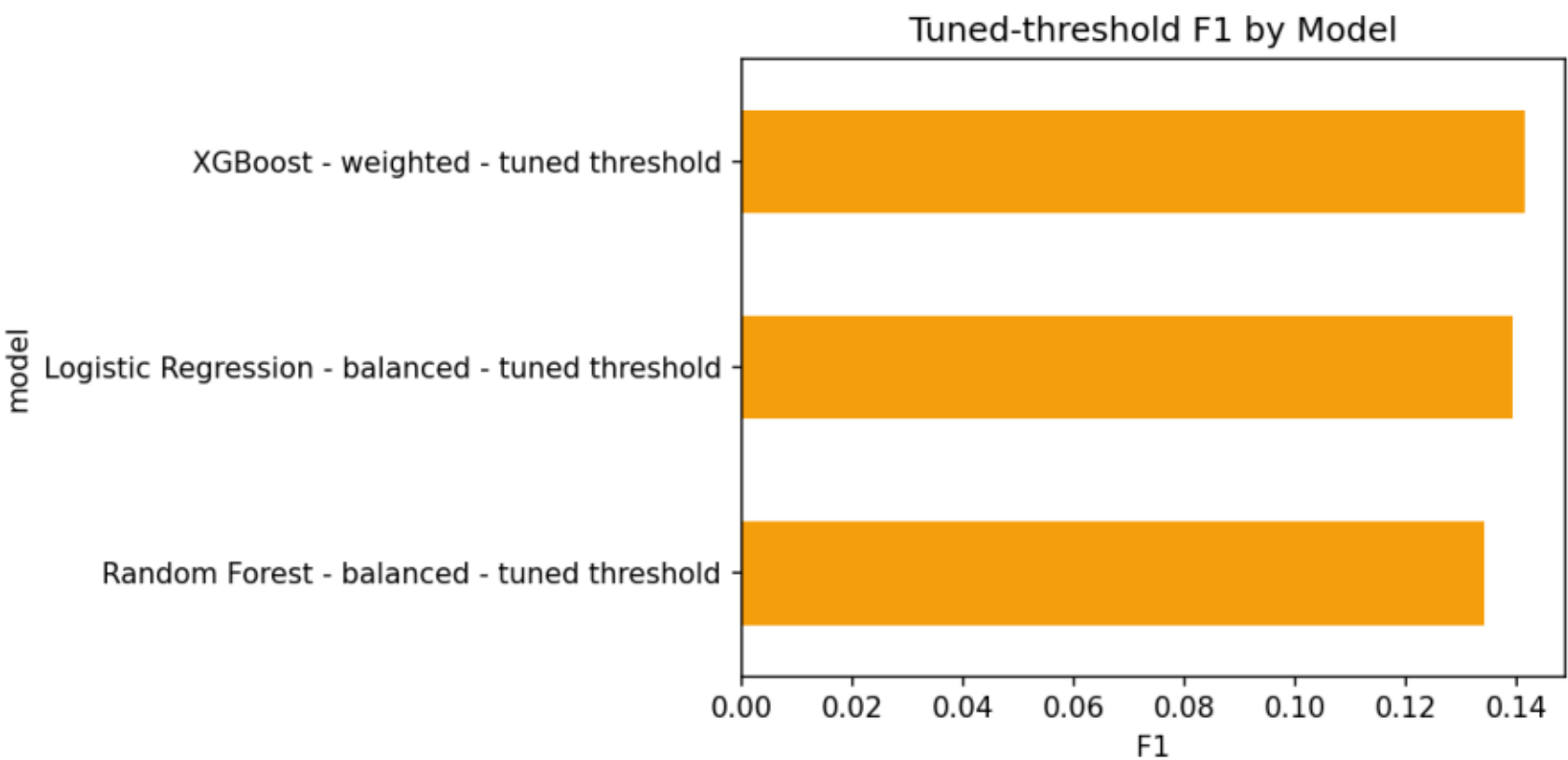
Target Distribution



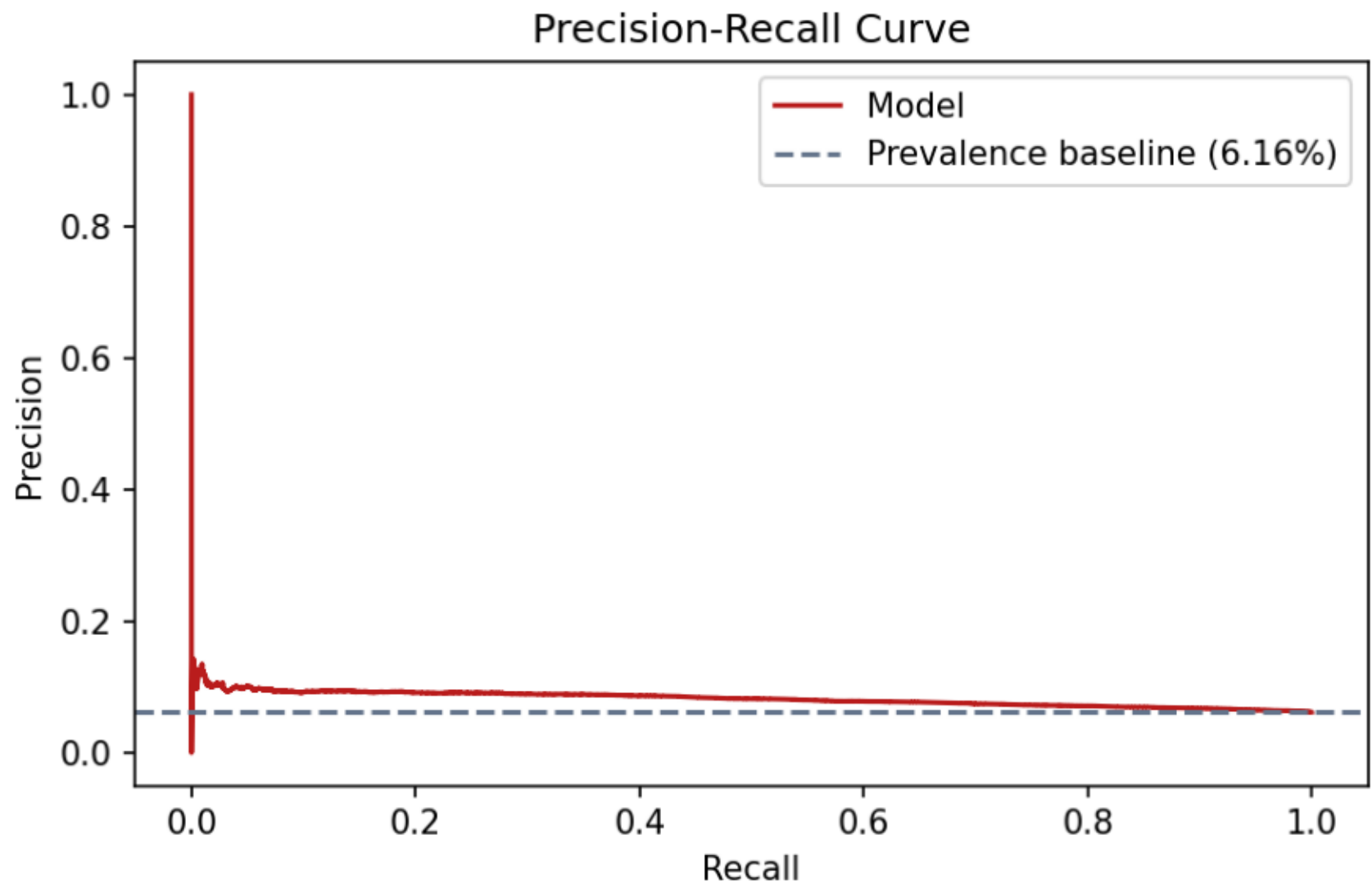
Feature Ablation Chart



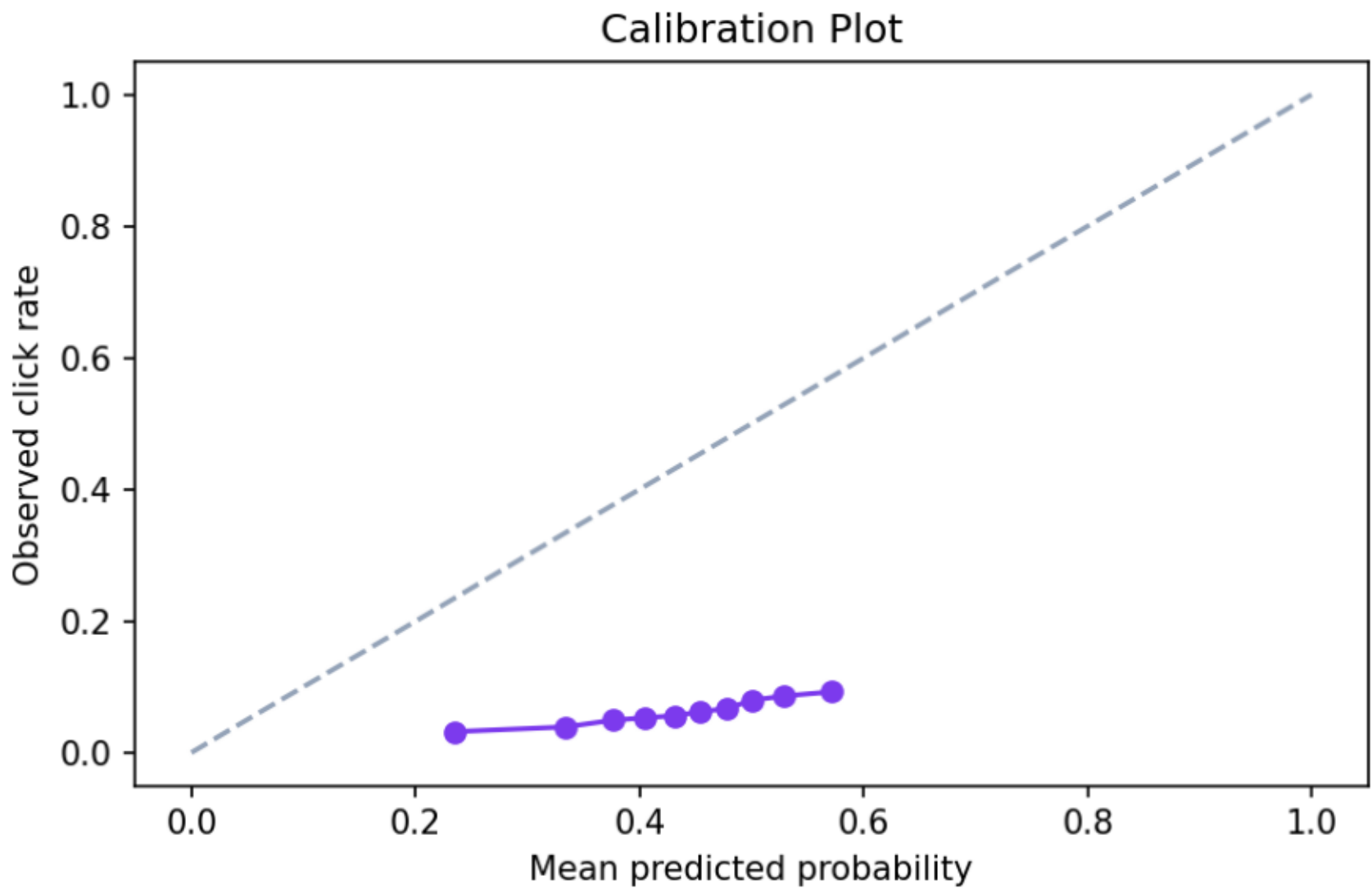
Model F1 Comparison



Precision-Recall Curve with Prevalence Baseline



Calibration Plot



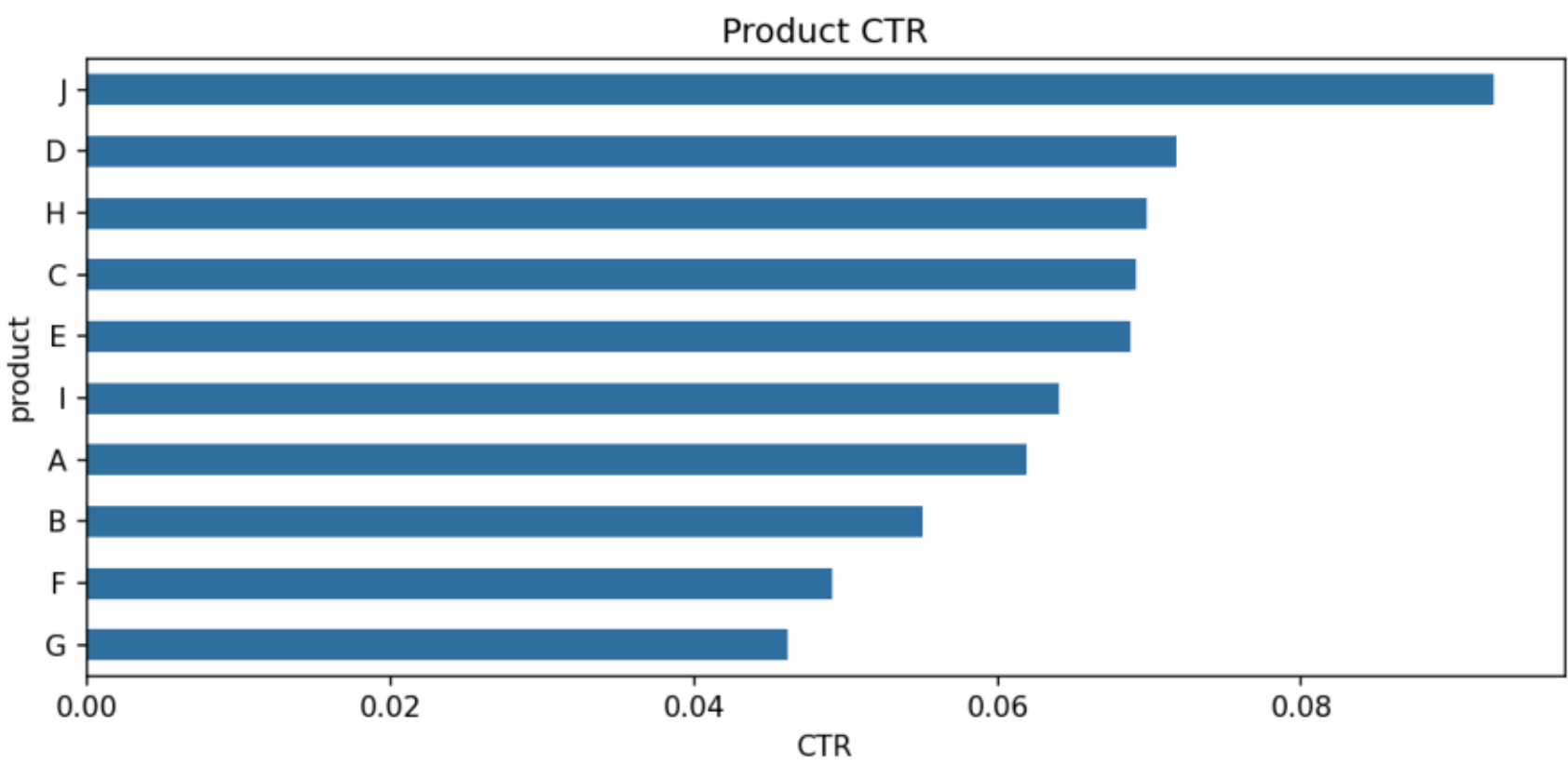
Production Lifecycle

Production Lifecycle for CTR Scoring

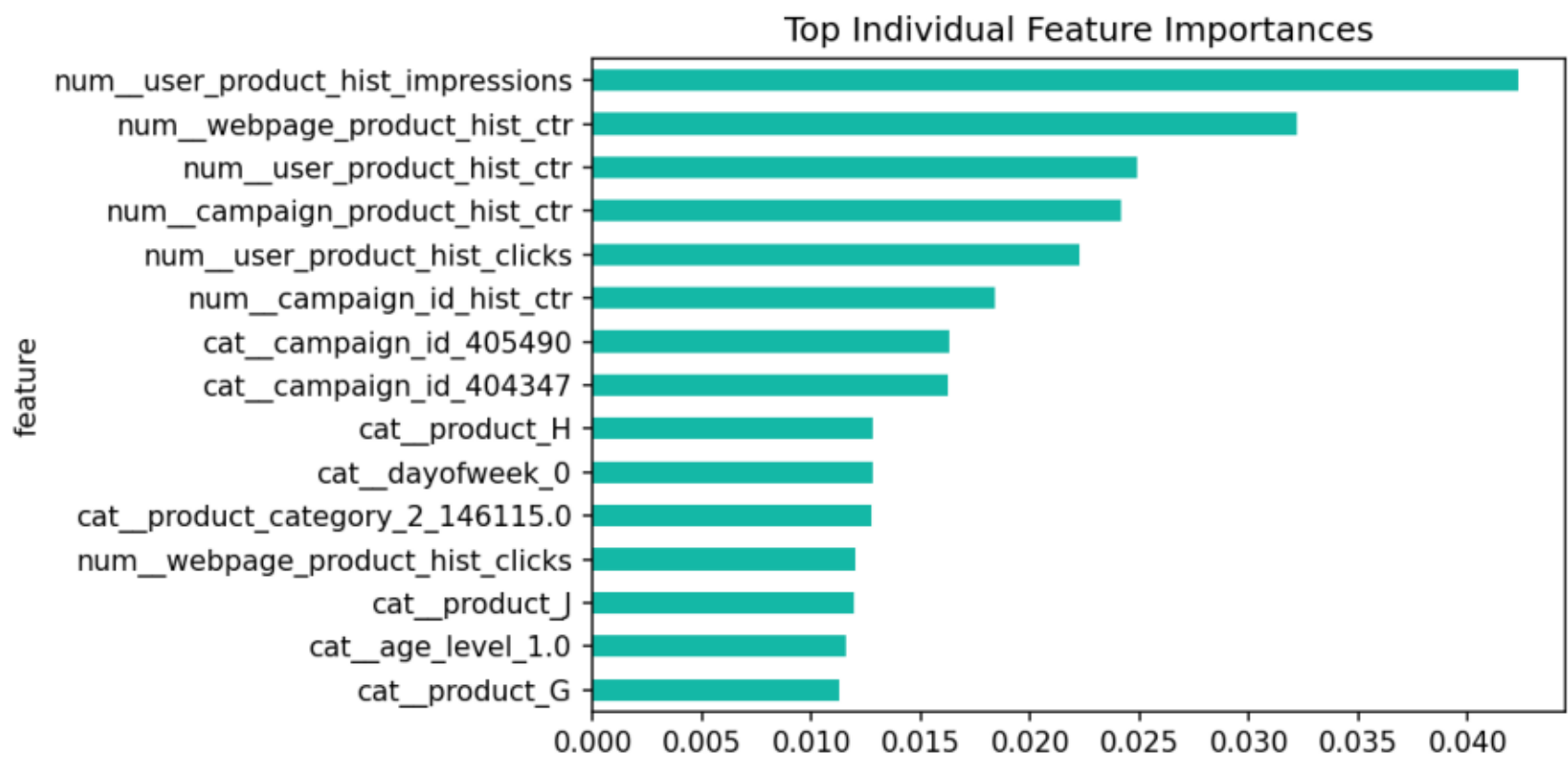


Monitor: PR-AUC ? Lift@K ? CTR ? calibration ? feature drift ? latency

Product CTR / Inventory Planning Signal



Top Individual Feature Importances



Cold-Start / Known-History Analysis

feature	segment	impressions	share_of_holdout	ctr	roc_auc	pr_auc
user_product	known_history	35923	0.5027	0.0495	0.5901	0.0673
user_product	cold_start	35543	0.4973	0.0738	0.5625	0.0882
campaign_product	known_history	71114	0.9951	0.0615	0.5937	0.0819
campaign_product	cold_start	352	0.0049	0.0682	0.5447	0.0949
webpage_product	known_history	71114	0.9951	0.0615	0.5937	0.0819
webpage_product	cold_start	352	0.0049	0.0682	0.5447	0.0949

Historical features are valuable, so cold-start behavior must be monitored separately.

Top Product CTR Planning Signal

product	clicks	impressions	ctr	expected_clicks_per_100k_impress
J	899	9698	0.0927	9,269.95
D	2949	41064	0.0718	7,181.47
H	7654	109574	0.0699	6,985.23
C	11306	163501	0.0691	6,914.94
E	1474	21452	0.0687	6,871.15
I	4079	63711	0.0640	6,402.35
A	953	15391	0.0619	6,191.93
B	1238	22479	0.0551	5,507.36
F	344	7007	0.0491	4,909.38
G	435	9414	0.0462	4,620.78

This is an inventory-planning signal, not a formal demand forecast.

High-Propensity User Profiles with Wilson Intervals

gender	age_level	city_development_index	clicks	impressions	ctr	ctr_ci_low	ctr_ci_high
Male	5.00	4.00	255	2945	0.0866	0.0770	0.0973
Female	5.00	4.00	100	1181	0.0847	0.0701	0.1019
Male	1.00	2.00	605	7922	0.0764	0.0707	0.0824
Female	5.00	2.00	208	2752	0.0756	0.0663	0.0861
Male	2.00	2.00	3736	50409	0.0741	0.0719	0.0764
Male	1.00	3.00	607	8291	0.0732	0.0678	0.0790
Male	1.00	4.00	306	4252	0.0720	0.0646	0.0801
Male	5.00	3.00	349	4909	0.0711	0.0642	0.0786
Male	2.00	3.00	1927	27671	0.0696	0.0667	0.0727
Male	5.00	2.00	493	7112	0.0693	0.0636	0.0755

ML Insight to Business Action

Model insight	Business action
High product CTR	Increase exposure and reserve more inventory
High webpage CTR	Prioritize placement and monitor page quality
High user-product historical CTR	Personalize ad selection and test targeted bid uplift
Weekend uplift	Test weekend bid modifier before permanent adoption
High predicted click propensity	Prioritize impression under budget constraints
Low CTR product	Refresh creative or reduce allocation
Model drift	Retrain and recalibrate thresholds

Recommendations are framed as tests/ranking actions because discrimination and calibration are moderate.

SMOTENC Conclusion and Limitations

? SMOTENC reduced false negatives by 21.7% in the controlled experiment, but F1 was nearly unchanged and PR-AUC decreased versus class weighting.

? Class weighting remains the simpler default unless the campaign explicitly values missed-click reduction more than training cost and ranking quality.

? Business-value thresholds are sensitivity analyses, not measured profit estimates, because campaign revenue and media-cost fields are unavailable.

? Apply Platt scaling or isotonic regression before using scores as literal expected click probabilities.

? Run A/B tests before turning offline lift into permanent bidding rules.